Mass Gap for the O(N) Model at Large N via Brascamp-Lieb and Feynman-Kac

M. R. Douglas

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Abstract

We outline a proof of the mass gap for the O(N) linear sigma model in two Euclidean dimensions at large N. The proof has three steps: (1) the gap equation determines the saddle point σ_* and the physical mass $m_0 = \sqrt{\sigma_*}$ (dimensional transmutation), (2) Brascamp-Lieb gives σ-concentration at σ_* with $O(1/N)$ variance, and (3) Feynman-Kac controls the $1/N$ corrections to the propagator mass. The convexity condition for Brascamp-Lieb, $\sigma_* > 1/(16\pi\lambda)$, is a condition on the renormalized mass vs. the coupling and is cutoff-independent.

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1 Setup

1.1 The O(N) linear sigma model

Let $\varphi = (\varphi^1, \ldots, \varphi^N) : \Lambda \to \mathbb{R}^N$ on a lattice $\Lambda \subset \mathbb{Z}^2$. The action is

$$S[\varphi] = \tfrac{1}{2} \sum_i \langle \varphi^i, (-\Delta)\varphi^i \rangle + N \sum_x \lambda \big(\tfrac{|\varphi(x)|^2}{N} - v^2 \big)^2. \tag{1}$$

The bare parameter v^2 is adjusted as a function of the UV cutoff Λ to keep the physical mass fixed (the only renormalization needed in 2d; the quartic coupling λ does not run).

1.2 The auxiliary field σ and the physical field u

Write $u(x) = |\varphi(x)|^2/N$ for the O(N)-invariant combination at each site. The interaction is $N\lambda(u(x) - v^2)^2$.

Definition A (pushforward). The pushforward of the interacting measure under u gives a probability measure μ_u on configurations $u : \Lambda \to [0, \infty)$. By the disintegration theorem:

$$\langle F(\varphi) \rangle = \int \mathbb{E}[F(\varphi) \,|\, u] \, d\mu_u. \tag{2}$$

Conditional on $u(x) = |\varphi(x)|^2/N$, each $\varphi(x)$ lives on a sphere $S^{N-1}_{\sqrt{Nu(x)}}$, coupled by the kinetic term. This is a well-defined positive measure, but the conditional expectation $\mathbb{E}[\varphi^i(x)\varphi^i(0) \,|\, u]$ is the propagator on the product of spheres, not exactly $(-\Delta + u)^{-1}(x, 0)$.

Definition B (HS auxiliary field). Introduce an auxiliary field σ at each site via the Gaussian identity:

$$e^{-N\lambda(u-v^2)^2} = c \int d\sigma \; e^{-N\lambda\sigma^2 + 2N\lambda\sigma(u-v^2)}. \tag{3}$$

Here $u = |\varphi(x)|^2/N$ is the same physical quantity as in Definition A, and σ is the new auxiliary variable that couples linearly to u. Applying (3) at each site, the φ-integral becomes Gaussian with covariance $(-\Delta + 4\lambda\sigma)^{-1}$. Integrating out φ exactly:

$$\langle \varphi^i(x)\varphi^i(0) \rangle = \mathbb{E}_\sigma\big[(-\Delta + 4\lambda\sigma)^{-1}(x, 0)\big] \tag{4}$$

where $d\mu_\sigma \propto \det(-\Delta + 4\lambda\sigma)^{-N/2} \, e^{-N\sum_x [\lambda\sigma^2 - 2\lambda\sigma v^2]} \, d\sigma$. This is **exact** at every N.

At the σ-saddle point: $\partial/\partial\sigma = 0$ gives $\sigma = u - v^2$, so the HS field measures the deviation of u from v^2.

Relation: Laplace transform. The pushforward density ρ_u (Definition A) and the HS determinant (Definition B) are related by

$$\mathbb{E}_{\mathrm{GFF}}\big[e^{-\alpha\cdot u}\big] = \frac{\det(-\Delta + 2\alpha/N)^{-N/2}}{\det(-\Delta)^{-N/2}}, \tag{5}$$

i.e., $\det(-\Delta+\tau)^{-N/2}$ is the moment generating function of $u = |\varphi|^2/N$ under the GFF, with $\tau = 2\alpha/N$. At $N = \infty$ (saddle-point of the inverse Laplace transform), the pushforward μ_u and the HS measure μ_σ both concentrate at $u_* = \sigma_* + v^2$, and both give the propagator $(-\Delta + u_*)^{-1}$. At finite N they differ by $O(1/N)$.

Convention. We use Definition B (HS auxiliary field) for the exact formula (4). Rescaling $\tau = 4\lambda\sigma$:

$$s_{\mathrm{eff}}[\tau] = \tfrac{1}{2}\mathrm{Tr}\log(-\Delta + \tau) + \sum_x V(\tau(x)), \tag{6}$$

where $V(\tau) = \tau^2/(16\lambda) - \tau v^2/2 + \mathrm{const}$, and (4) becomes $\langle\varphi^i\varphi^i\rangle = \mathbb{E}_\tau[(-\Delta+\tau)^{-1}]$.

2 The gap equation

2.1 Saddle point of s_{eff}

For uniform $\sigma(x) = \sigma_0$, the stationarity condition $ds_{\mathrm{eff}}/d\sigma_0 = 0$ is

$$\tfrac{1}{2}G(x,x;\sigma_*) = 2\lambda(v^2 - \sigma_*), \tag{7}$$

where $G(x,x;\sigma_0) = \int \frac{d^2k}{(2\pi)^2}\frac{1}{k^2+\sigma_0}$. Note that v^2 is *not* a stationary point: at $\sigma_0 = v^2$ the derivative $\frac{1}{2}G(x,x;v^2) > 0$, so the minimum lies at $\sigma_* < v^2$.

In 2d with cutoff Λ: $G(x,x;\sigma_*) \approx \frac{1}{4\pi}\log\frac{\Lambda^2}{\sigma_*}$, giving

$$\frac{1}{8\pi}\log\frac{\Lambda^2}{\sigma_*} = 2\lambda(v^2 - \sigma_*). \tag{8}$$

For $\sigma_* \ll v^2$: $\sigma_* \approx \Lambda^2 e^{-16\pi\lambda v^2}$, exponentially small (dimensional transmutation).

2.2 Physical mass and renormalization

The physical mass at $N = \infty$ is $m_0 = \sqrt{\sigma_*}$, since $\sigma = \sigma_*$ deterministically and $(-\Delta + \sigma_*)^{-1}(x,0) \sim e^{-m_0|x|}$.

In the continuum limit ($\Lambda \to \infty$), we adjust the bare parameter:

$$v^2(\Lambda) = \sigma_* + \frac{1}{16\pi\lambda}\log\frac{\Lambda^2}{\sigma_*} \tag{9}$$

to keep σ_* (and hence m_0) fixed. The quartic coupling λ does not run (super-renormalizable in 2d).

3 Brascamp-Lieb concentration

3.1 Hessian at σ_*

The Hessian of s_{eff} at σ_*, in Fourier, is

$$H(k) = 2\lambda - \tfrac{1}{2}\Pi(k), \quad \Pi(k) = \int \frac{d^2p}{(2\pi)^2} \frac{1}{(p^2+\sigma_*)((k-p)^2+\sigma_*)}. \tag{10}$$

At $k = 0$: $\Pi(0) = 1/(4\pi\sigma_*)$, so $H(0) = 2\lambda - 1/(8\pi\sigma_*)$.

3.2 Convexity condition

$H(k)$ is positive definite for all k when $H(0) > 0$, i.e.,

$$\boxed{\sigma_* > \frac{1}{16\pi\lambda}.} \tag{11}$$

This is a condition on the **renormalized** parameters σ_* and λ. It is **cutoff-independent**: since λ does not run and σ_* is held fixed by renormalization, the condition (11) either holds or fails independently of Λ. In particular, it survives the continuum limit.

Equivalently: the physical mass must satisfy $m_0 > 1/\sqrt{16\pi\lambda}$. For $\lambda = 1$: $m_0 > 0.14$.

3.3 Brascamp-Lieb bounds

When (11) holds, s_{eff} is strictly convex at σ_* with convexity parameter $\kappa = 2\lambda - 1/(8\pi\sigma_*) > 0$. The measure $e^{-Ns_{\text{eff}}}$ is log-concave, and Brascamp-Lieb [1] gives:

$$\mathrm{Cov}(\sigma(x), \sigma(y)) \le \frac{1}{N} H^{-1}_{xy}, \tag{12}$$

$$\mathrm{Var}(\sigma(x)) \le \frac{1}{\kappa N}, \tag{13}$$

$$|\kappa_k(\sigma(x))| \le \frac{C^k\, k!}{(\kappa N)^{k-1}}, \quad k \ge 2. \tag{14}$$

The σ-field concentrates at σ_* with $O(1/\sqrt{N})$ fluctuations.

4 Exponential decay of the propagator

4.1 Feynman-Kac representation

The propagator at fixed σ has the path integral representation:

$$(-\Delta + \tau)^{-1}(x, 0) = \int_0^\infty ds\, \mathbb{E}^W_{0\to x,s}\Big[e^{-\int_0^s \tau(\omega(t))\,dt}\Big] \tag{15}$$

where $\tau = 4\lambda\sigma$ and $\mathbb{E}^W$ is over lattice random walks from 0 to x in time s.

4.2 Sub-Gaussian bound for log-concave measures

The σ-measure $e^{-Ns_{\text{eff}}}$ is log-concave (when (11) holds). For log-concave random vectors, every linear functional is sub-Gaussian (Borell's lemma):

$$\mathbb{E}_\sigma[e^X] \le \exp\bigl(\mathbb{E}[X] + \tfrac{1}{2}\mathrm{Var}(X)\bigr) \tag{16}$$

for any linear functional X of σ.

Applied to $X = -\int_0^s \tau(\omega(t))\,dt$ (linear in σ for fixed path ω):

$$\mathbb{E}[X] = -s\,\tau_* \quad (\text{where } \tau_* = 4\lambda\sigma_*), \tag{17}$$

$$\mathrm{Var}(X) = \int_0^s\!\!\int_0^s \mathrm{Cov}\bigl(\tau(\omega(t_1)), \tau(\omega(t_2))\bigr)\,dt_1\,dt_2 \le \frac{s}{N}\,\|C_\tau\|_1, \tag{18}$$

where $\|C_\tau\|_1 = \int \mathrm{Cov}(\tau(0), \tau(z))\,dz \le (4\lambda)^2 \int H_{0z}^{-1}/N\,dz$ is the integrated τ-covariance (finite, from the BL exponential decay of H^{-1}).

4.3 The key bound

Combining (15), (16), (17), (18):

$$\begin{aligned}\mathbb{E}_\sigma[(-\Delta+\tau)^{-1}(x,0)] &\le \int_0^\infty ds\; \mathbb{E}^W_{0\to x,s}\bigl[e^{-s\tau_* + s\|C_\tau\|_1/(2N)}\bigr] \\ &= \boxed{(-\Delta+m^2)^{-1}(x,0)}\end{aligned} \tag{19}$$

where

$$m^2 = \tau_* - \frac{\|C_\tau\|_1}{2N} = 4\lambda\sigma_* - \frac{8\lambda^2}{N}\int H_{0z}^{-1}\,dz. \tag{20}$$

The averaged propagator is bounded by a **deterministic** Green's function $(-\Delta+m^2)^{-1}$, which decays as $e^{-m|x|}$ by standard lattice Fourier analysis.

4.4 Mass gap condition

The mass $m > 0$ when $m^2 > 0$, i.e.,

$$N > N_0 := \frac{2\lambda \int H_{0z}^{-1}\,dz}{\sigma_*}. \tag{21}$$

The threshold N_0 is finite and computable from the lattice σ-propagator H^{-1}.

4.5 Main theorem

Theorem 1 (Mass gap at large N)**.** *For the O(N) LSM on $\mathbb{Z}^2$ with $N \ge 3$, $\sigma_* > 1/(16\pi\lambda)$ (BL convexity), and $N > N_0$ (from* (21)*), the connected two-point function decays exponentially:*

$$|\langle\varphi^i(x)\varphi^i(0)\rangle_c| \le C\,e^{-m|x|} \tag{22}$$

with $m > 0$ given by (20)*.*

Proof. 1. **HS** (§1.2): The joint measure on (σ, φ) factors as Gaussian$(\varphi|\sigma) \times \mu_\sigma$, giving the exact formula (4). The σ-measure $\mu_\sigma \propto \det(-\Delta + \tau)^{-N/2} e^{-N \cdot \text{quartic}}$ is well-defined (Nelson bound, same as for the interacting measure).

2. **Gap equation** (§2): The minimum σ_* of s_{eff} satisfies (7), with $\sigma_* > 0$ always in 2d.

3. **BL** (§3): At σ_*, the Hessian H is positive (by (11)), and $\text{Cov}(\sigma(x), \sigma(y)) \leq H^{-1}_{xy}/N$ with exponential decay.

4. **Sub-Gaussian bound**: $X = -\int \tau(\omega)\, dt$ is a linear functional of the log-concave σ, hence sub-Gaussian by Borell's lemma.

5. **FK bound**: The averaged propagator $\leq$ deterministic $(-\Delta + m^2)^{-1}$ with m^2 from (20).

6. **Mass gap**: $m^2 > 0$ when $N > N_0$. Exponential decay of $(-\Delta + m^2)^{-1}$ is standard. □

5 Discussion

5.1 The convexity condition is physical

The condition $\sigma_* > 1/(16\pi\lambda)$ is equivalent to $m_0^2 > 1/(16\pi\lambda)$, a lower bound on the physical mass. It is cutoff-independent and survives the continuum limit, since λ does not run in 2d and σ_* is the renormalized mass squared.

For strong coupling (λ large): the threshold $1/(16\pi\lambda) \to 0$, so the condition is easily satisfied. For any fixed $\lambda > 0$: the condition holds for $m_0 > 1/\sqrt{16\pi\lambda}$.

5.2 No cluster expansion needed

Unlike Kupiainen [2], who uses a cluster expansion to control the $1/N$ corrections for the NLSM, our proof uses only:

- The gap equation (a single self-consistent equation),
- Brascamp-Lieb concentration (a convexity-based inequality),
- Feynman-Kac cumulant expansion (path integral representation).

The key simplification: the convexity of s_{eff} at σ_* (when (11) holds) makes the σ-measure log-concave, replacing the cluster expansion with BL.

5.3 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). The σ-field cannot detect angular topology. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

5.4 Comparison with Kupiainen

	This work	Kupiainen [2]
Model	LSM (finite λ)	NLSM ($\lambda = \infty$)
σ-control	Brascamp-Lieb	Cluster expansion
φ-decay	Feynman-Kac	Random walk expansion
Mass formula	$m = \sqrt{\sigma_*}(1 + O(1/N))$	$m = m_0(1 + O(1/N))$
Continuum limit	Yes (renormalized)	Yes

Since the LSM and NLSM share the same infrared physics (they differ only at scale $M_\sigma \sim \sqrt{\lambda}\, v$), both proofs establish the same physical mass gap in the large-N limit.

References

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